

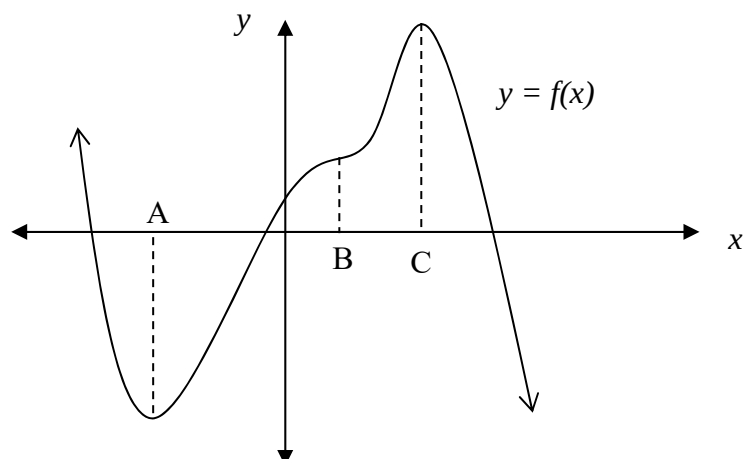
QUESTION 1 (12 Marks) **Marks**

- (a) Consider the parabola $(x-4)^2 = 8(y+3)$.
- (i) State the co-ordinates of the vertex **1**
 - (ii) Find the focal length. **1**
 - (iii) Find the co-ordinates of the focus **1**
 - (iv) Find the equation of the directrix. **1**
- (b) Re-write $y = x^2 + 4x + 3$ in the form $(x-h)^2 = 4a(y-k)$. **2**
- (c) A parabola has its focus at the point $(2, -1)$ and directrix $y = 3$.
- Find
- (i) the focal length. **1**
 - (ii) the co-ordinates of the vertex. **1**
 - (iii) the equation of the parabola. **2**
- (d) Sketch the locus of the point $P(x,y)$, where P is 3 units from the point $A(3,1)$ and hence write down its equation, **2**

QUESTION 2 (14 Marks) Start this question on a new page.

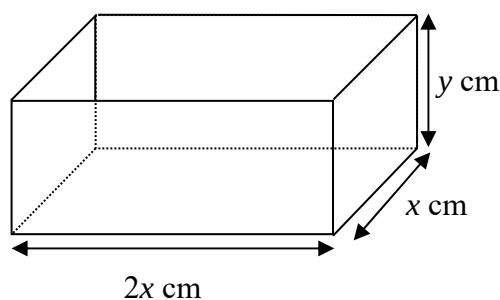
- (a) Consider the curve $y = \frac{x^3}{3} - 4x$
- (i) Show that $\frac{dy}{dx} = x^2 - 4$ **1**
 - (ii) Find the points where the curve crosses the axes. **2**
 - (iii) Find the coordinates of any stationary points and determine their nature. **3**
 - (iv) Find any points of inflection. (Change in concavity must be shown.) **2**
 - (v) Draw a graph of this function (about $\frac{1}{2}$ page.) **2**
 - (vi) Find the equation of the tangent to this curve at the point $(3,-3)$. **2**
- (b) Sketch the graph of $y = f(x)$ such that **2**
 $f(3) = 5$, $f'(3) = 0$, $f'(x) > 0$ for $x < 3$ and $f'(x) < 0$ for $x > 3$.

Continued page 2

QUESTION 3**(12 marks)**Start this question on a new page.**Marks****(a)****3**

Draw or trace the graph $y = f(x)$, above, on your writing paper. On the same diagram, draw a graph of $y = f'(x)$, the derivative of the function, indicating the points A, B and C.

- (b) An open rectangular box has four sides and a base, but no lid, as in the figure below.



The dimensions of the base of the box are x cm by $2x$ cm and the height is y cm.

- (i) Write down the formula for the external surface area A cm² of the box in its simplest form. **2**
- (ii) Write down the formula for the volume V cm³ contained by the box in its simplest form. **2**
- (iii) Given that the surface area, A , of the box is 150 cm², show that the formula for the volume in terms of x is $V = \frac{150x - 2x^3}{3}$. **2**
- (iv) Find the value of x for which V is a maximum, and verify that the maximum value of V is $\frac{150}{3}$ cm³. **3**

End of Paper.

11 Mathematics 2007 Yearly

QUESTION 1

(a)(i) $(x-4)^2 = 8(y+3)$

So ~~x~~ vertex:

$$x-4=0$$

$$x=4$$

$$y+3=0$$

$$y=-3$$

$\therefore$ Vertex at $(4, -3)$ ①

(ii) $4a = 8$
 $a = 2$

$\therefore$ focal length is 2 ①

(iii) focus is a units above vertex (concave up)

$$\therefore \text{focus is } (4, (-3+2))$$
$$= (4, -1) \quad \text{①}$$

(iv) directrix is a units below vertex

$$\therefore \text{directrix is } y = -3 - 2$$

$$\text{i.e. } y = -5 \quad \text{①}$$

(b) $y = x^2 + 4x + 3$

$$\text{So } x^2 + 4x + 4 = y + 1$$

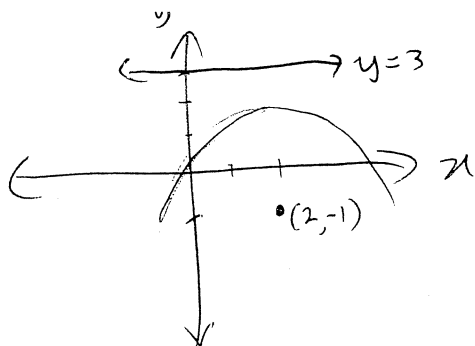
$$(x+2)^2 = y+1$$

①
for
LHS

$$= 4\left(\frac{1}{4}\right)(y+1)$$

① for RHS

(c)



(i) Distance from focus to directrix is $2a$

$$\text{So } 2a = 3 - (-1) \\ = 4$$

$$a = 2$$

$\therefore$ focal length is 2. ①

(ii) The parabola is concave down.

The vertex halfway between the focus and directrix.

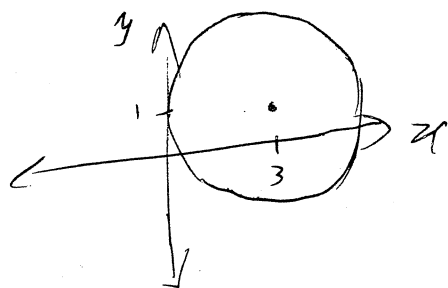
$$\text{i.e. at } \left(2, \frac{-1+3}{2}\right) \quad \text{or} \quad \text{focus plus a units up.} \\ = (2, 1) \quad \text{①}$$

$$(iii) (x-h)^2 = 4a(y-k)$$

sub. in vertex and focal length.

$$\left. \begin{aligned} (x-2)^2 &= -8(y-1) \\ x^2 - 4x + 8y - 4 &= 0 \end{aligned} \right\} \text{either ②} \quad \begin{array}{l} -1 \text{ mark if minus} \\ \text{sign left out for} \\ \text{concave down} \end{array}$$

(d)



① circle must touch y-axis

$$\left. \begin{aligned} (x-3)^2 + (y-1)^2 &= 9 \\ x^2 - 6x + y^2 - 2y + 1 &= 0 \end{aligned} \right\} \text{either ①}$$

QUESTION 2

a) i) $y = \frac{x^3}{3} - 4x$

$\therefore y' = \frac{3x^2}{3} - 4$

$y' = x^2 - 4 \quad (1)$

ii) $\frac{x^3}{3} - 4x = 0$

$x(\frac{x^2}{3} - 4) = 0$

$\therefore$ Crosso at $-2\sqrt{3}, 0, 2\sqrt{3}$

$\therefore x \geq 0$ or $\frac{x^2}{3} - 4 = 0$

(1)

$x^2 = 12$

$x = \pm 2\sqrt{3}$
(1)

iii) Stationary Points when $y' = 0$

$\therefore x^2 - 4 = 0$

$x^2 = 4$

$x = \pm 2 \quad (1)$

Also $y'' = 2x$

$\therefore y''(2) = 4$

> 0

$\therefore$ Min

$y''(-2) = -4$

< 0

$\therefore$ Max (1)

$\therefore (2, -5\frac{1}{3})$ is a minimum turning point

$(-2, 5\frac{1}{3})$ is a maximum turning point (1)

iv) Points of inflexion when $y'' = 0$

$\therefore 2x = 0$

$x = 0 \quad (1)$

$x \quad -1 \quad 0 \quad 1$

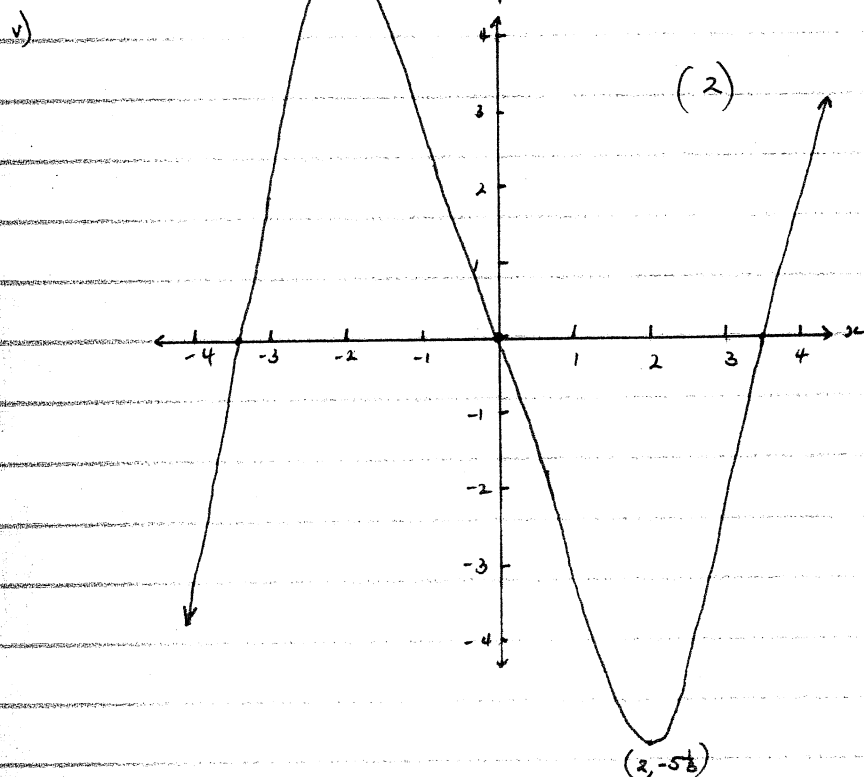
$f''(x) \quad -2 \quad 0 \quad 2$

< 0

$> 0 \quad (1)$

$\therefore$ Since concavity changes

$(0, 0)$ is a point of inflexion



vi) Given $(3, -3)$

$y' = x^2 - 4$ when $x = 3$

$y' = 9 - 4$

$y' = 5 \quad (1)$

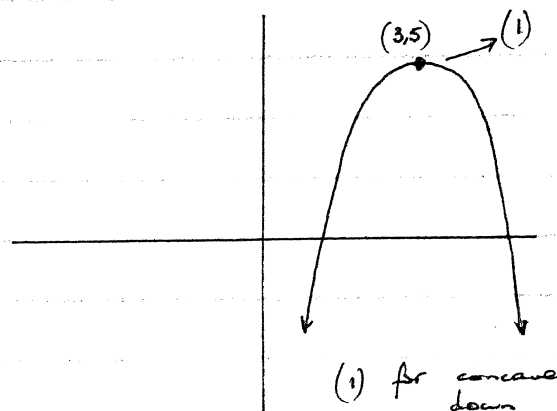
$\therefore m = 5 \quad (3, -3)$

$y + 3 = 5(x - 3)$

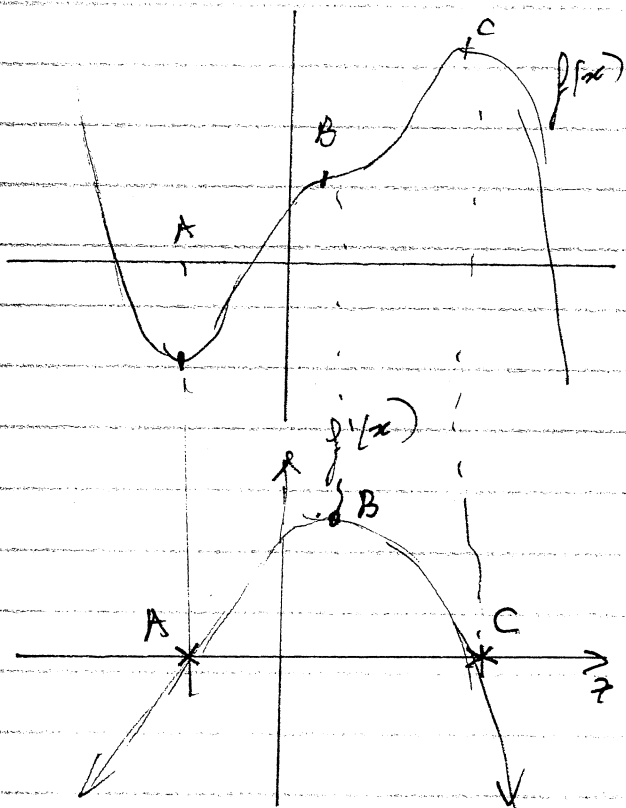
$y + 3 = 5x - 15$

$y = 5x - 18 \quad (1)$

b)



QUESTION THREE



(11)

Given $A = 150 = 6xy + 2x^2$

$$\therefore 6xy = 150 - 2x^2$$

$$y = \frac{150 - 2x^2}{6x}$$

$$= \frac{2(75 - x^2)}{3x}$$

$$y = \frac{75 - x^2}{3x}$$

Sub into V

$$V(x, y) = 2x^2 y$$

$$V(x) = 2x^2 \left[\frac{75 - x^2}{3x} \right]$$

$$= \frac{2x}{3} [75 - x^2]$$

$$V(x) = \frac{150x - 2x^3}{3}$$

QED.

(b) Surface Area =

$$2x(x, y) + 2(2x \cdot y) + (2x \cdot x)$$

$\xrightarrow{\text{Sides}}$ $\xrightarrow{\text{Front/Back}}$ $\xrightarrow{\text{bottom}}$

$$A(x, y) = 2xy + 4xy + 2x^2$$

$$A(x, y) = 6xy + 2x^2$$

(ii)

Volume = $L \times w \times h$

$$V(x, y) = (2x) \cdot (x) \cdot (y)$$

$$V(x, y) = 2x^2 y$$

For MAX/MIN Find $V'(x) = 0$

$$V(x) = \frac{150x}{3} - \frac{2x^3}{3}$$

$$V'(x) = \frac{150}{3} - 2 \cdot \frac{1}{3} x^2$$

$$V'(x) = \frac{150}{3} - 2x^2 = 0$$

For MAX/MIN

$$\therefore 2x^2 = \frac{150}{3}$$

$$x^2 = \frac{25}{3} \quad x^2 = 25$$

$$x = \pm \frac{5\sqrt{3}}{3} \quad x = \pm 5$$

Since $x > 0$

$$\therefore x = \frac{5\sqrt{3}}{3} \quad 5$$

$$V\left(\frac{5\sqrt{3}}{3}\right) = \frac{150}{3} \cdot \left(\frac{5\sqrt{3}}{3}\right) - \frac{2}{3} \left(\frac{5\sqrt{3}}{3}\right)^3$$

$$= \frac{150}{3}$$

$$V(5) = \frac{150(5) - 2(5)^3}{3}$$

$$= \frac{500}{3}$$